

# Development and validation of virtual prism adaptation therapy

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**Abstract**—Hemispatial neglect is common after stroke and associated with poor functional recovery and social return. Among various possible treatments for neglect, prism adaptation therapy has been known to be effective to promote the recovery from neglect. However, the conventional prism therapy using a prism glass and real object has a few shortcomings. Prism lens has to be changed to adjust the degree of deviation by the prism and the hand trajectory has to be masked during therapy. To solve these problems in conventional prism therapy, we developed virtual prism adaptation therapy using immersive virtual reality and depth-sense camera. This system was applied to 4 healthy people to validate it. The experiment consisted of 4 phases, which were non-prism, 5-min-prism (10-degree deviation), 5-min-prism (20-degree deviation) and 5-min-non-prism (post-adaptation). The median values of pointing deviations for 30-second-block in each phase were plotted. Deviations to right side were observed during prism phase and left side deviations were observed during the post-adaptation phase, which is similar phenomenon compared to the conventional prism therapy. The virtual prism adaptation therapy could be a low-cost system to solve the problems in conventional prism adaptation therapy. Further research in more healthy people and patients is required.

**Keywords**—neglect; virtual reality; prism; adaptation; rehabilitation; stroke

## I. INTRODUCTION

Hemispatial neglect is common after stroke and associated with poor functional recovery and social return[1-3]. Therefore, the effort to rehabilitate the neglect is important to reduce the disability and socioeconomic burden after stroke. However, rehabilitation for neglect is often neglected in the clinical practice[4].

For instance, prism adaptation therapy is the modality for neglect, which has shown the highest clinical evidence from the previous studies[5,6]. However, the conventional prism therapy using a prism glass and real object has a few shortcomings. Prism lens has to be changed to adjust the degree of deviation by the prism and the hand trajectory has to be masked during therapy.

To solve the above-mentioned problems, we developed virtual prism adaptation therapy using immersive virtual reality and depth-sense camera.

## II. MATERIALS AND METHODS

### A. Virtual Prism Adaptation System(Fig. 1)

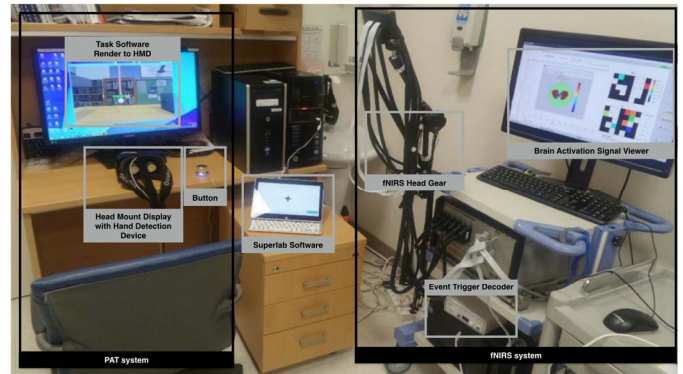


Fig. 1 Virtual prism adaptation system

In virtual prism adaptation system, Oculus Rift DK 2 and Leap Motion controller were used to visualize virtual reality and control head movement. Functional near infrared spectroscopy (fNIRS) system was used to measure the brain activity through a connection with the Super Lab software which transfers triggers to the system. PAT system and fNIRS system with Super lab software was synchronized by using TCI/IP communications. The fNIRS system will be used for the further work to analyze the cortical activation area related with the virtual prism adaptation task.

The visualized virtual hand was rendered in the transformed position as deviation angle to make prism effect. So, in deviated case, positions and trajectories of real hand and virtual hand is different (Fig. 2)

### B. Subjects and Experiment Design

This system was applied to 4 healthy people to validate it. The experiment consisted of 4 phases, which were non-prism, 5-min-prism (10-degree deviation), 5-min-prism (20-degree deviation) and 5-min-non-prism (post-adaptation). Each phase was composed of 30-second-block and subjects were instructed to point the target object in the immersive virtual reality in every 3 seconds (Fig. 2). The median values of pointing

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deviations for 30-second-block in each phase were plotted.

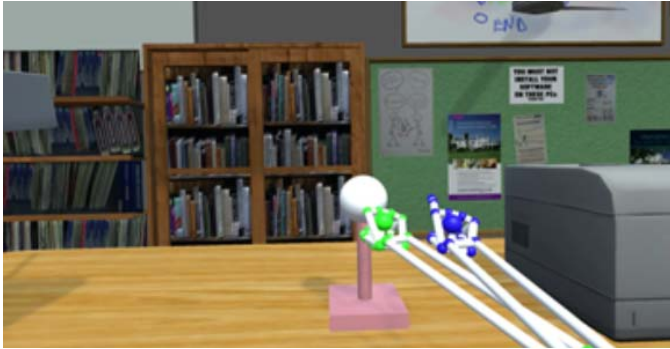


Fig. 2 Hand trajectory in the virtual reality. Green hand is the real hand trajectory and blue hand is the deviated hand trajectory in the virtual reality by the virtual prism adaptation system.

### III. RESULTS

Deviations to right side were observed during prism phase and left side deviations were observed during the post-adaptation phase, which is similar phenomenon compared to the conventional prism therapy (Fig. 3).

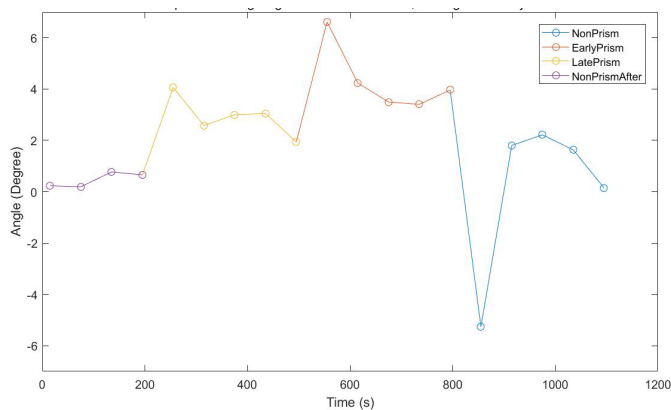


Fig. 3 Graph of pointing error (angle between the target and middle finger) in time series data.

### IV. CONCLUSIONS

The virtual prism adaptation therapy could be a low-cost system to solve the problems in conventional prism adaptation therapy. In the further work, to measure brain activity in virtual prism adaptation task, the fNIRS system will be used. Further research in more healthy people and patients is required.

### ACKNOWLEDGMENT

Won-Seok Kim, Sungmin Cho and Nam-Jong Paik have a patent pending entitled 'Method, system and readable recording medium of creating visual stimulation using virtual model', the number is "10-2016-0183569", which is relevant to this work.

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